SYSTEM OVERVIEW

Enterprise AI Customer Operations

An operator-led customer-case system combining durable event processing, prompt version control, human review and measured evaluations. The application was verified locally in deterministic demo mode, not against a live external language model.

Implemented architecture

React / Vite / TypeScript supplies Overview, Cases, Prompt Studio, Evaluations and Operations. A typed **Express 5** API validates requests and persists state using **Prisma 6 / PostgreSQL**. A separate Node process publishes outbox events to **RabbitMQ** and consumes work. The background process invokes either a deterministic demo provider or a configured OpenAI-compatible Fetch endpoint.

Case submitted	Atomic database write	Confirmed broker delivery	Independent worker	Review and resolution

Data ownership and delivery guarantees

A single PostgreSQL transaction records a case, timeline entry, processing job and outbox event. Publisher confirms and manual RabbitMQ acknowledgements implement **at-least-once** delivery; worker leases and unique persisted results stop duplicate committed outcomes. Provider calls can repeat after a crash, so this is not exactly-once inference.

Failed jobs use bounded retries with delayed attempts and dead-letter archives. The Operations view exposes attempts, worker heartbeat and explicit replay. Replay creates an auditable successor job rather than replacing its original history.

Human decisions and prompt versions

Classification and response recommendations use separate prompt templates. Jobs capture exact prompt version IDs at enqueue time. Published or evaluated versions become immutable, and activation is audited. Operators can correct classification or draft text, approve suggestions and resolve cases; **no generated customer response is automatically sent**.

Safety and provider boundary

Zod validates classification and response output. Security, immediate safety and widespread outages route to Critical human review regardless of model suggestions. Customer input is segregated from system policy, but this is not a demonstrated prompt-injection security guarantee. The demo provider uses deterministic rules, never a model call; real-provider behavior and Azure Service Bus connectivity were not live-verified.

VERIFICATION & LIMITATIONS

What was actually tested

Reported results come from the repository's local verification record dated 14 October 2026. These checks establish application behavior in the specified environment; they do not measure production AI performance.

Executed checks

17 passed	Backend unit and messaging-adapter tests, including provider contracts, schema validation, policy rules and broker stubs.
6 passed	Frontend workflow tests including forms, API failures, selected prompts and navigation.
9 passed	Integration tests using real PostgreSQL and RabbitMQ: duplicates, retries, dead-lettering, replay, leases, human reviews and prompt/evaluation flows.
Verified	Both production builds, type checks and lint; Docker Compose API/database/broker/worker startup; browser case workflow and refresh persistence.

Prompt evaluation

A seed dataset contains **36 labelled synthetic requests**. Evaluation jobs compute category and priority accuracy, per-class precision/recall/F1, six-class macro-F1, schema compliance and attempted inference latency. Errors are retained; token counts and costs stay null when unavailable. Comparisons show metric deltas and example regressions.

Only completed **real-provider** evaluations meeting configured thresholds qualify for the normal activation gate. Demo runs need an explicit override reason. Demo scores establish pipeline correctness, not externally validated LLM quality.

Not yet validated for production

External OpenAI-compatible inference and Azure Service Bus were exercised through adapters and stubs, not live accounts. GitHub Actions exists but did not run remotely during the documented session. The single-operator token is not multi-user identity, roles or tenant isolation. There is no automatic customer delivery, production deployment, verified external-model quality or full operational monitoring.

Key repository evidence

docs/version-1-report.md, **docs/testing.md**, **docs/architecture.md**, **docs/ai-llmops.md**, **docs/messaging.md**; backend **src/services/processing.ts**, **src/messaging/outbox.ts**, **src/evaluation/scoring.ts**, **prisma/schema.prisma**; frontend **src/App.tsx**.

Source: github.com/keenosmith-del/enterprise-ai-customer-operations